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Semester	B.E. Semester VII – INFT
Subject	WAS
Subject Professor In-charge	Prof Rakshad Sood

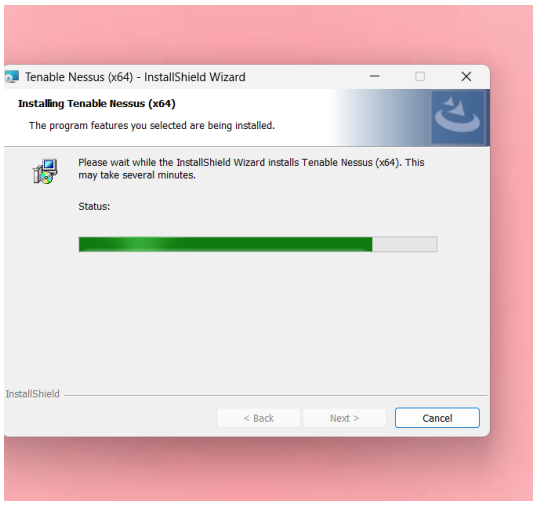
Student Name	Chirag Sharma
Roll Number	24101C2007
Assignment No	**06**
Assignment Title	**Nessus Vulnerability Scanner**
Description	**Aim** To perform vulnerability assessment on a target system using Nessus Vulnerability Scanner and identify security vulnerabilities, misconfigurations, and potential risks.
Procedure	**Theory** Nessus is a vulnerability assessment tool developed by Tenable. It scans systems, networks, and applications to identify known security weaknesses. It checks for: • Missing security patches • Outdated software • Open ports and running services • Weak configurations • Known vulnerabilities (CVEs) • SSL/TLS configuration issues Nessus provides a detailed report containing the vulnerability name, severity, description, and recommended solution. **Severity Levels** **Severity** — **Description** Critical — Vulnerabilities that may allow severe system compromise High — Serious vulnerabilities requiring prompt attention Medium — Vulnerabilities with moderate risk Low — Minor security weaknesses Informational — Useful system or configuration information **Requirements**

- Windows/Linux system
- Nessus Essentials installed
- Internet connection for activation and plugin updates
- A target machine that you own or have permission to scan

Procedure

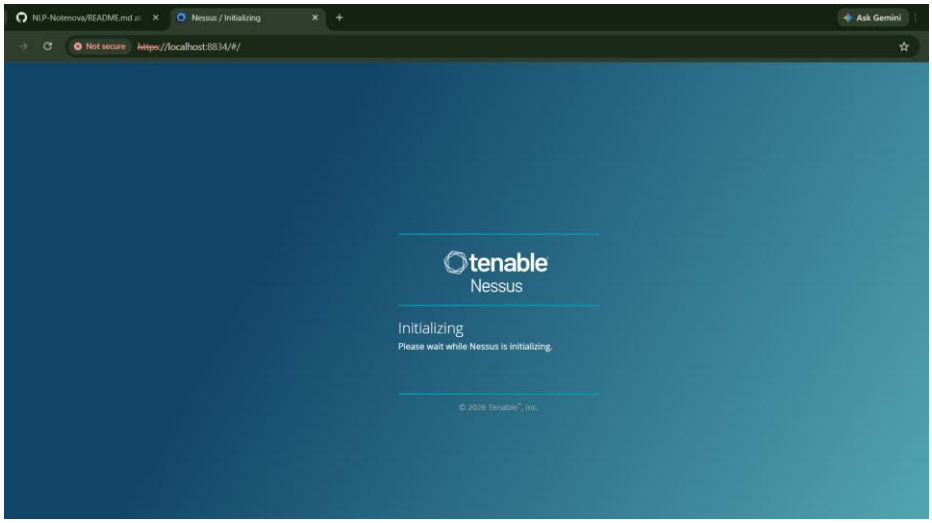
Step 1: Install Nessus

1. Download Nessus Essentials from the official Tenable website.
2. Install Nessus according to your operating system.



3. Start the Nessus service.
4. Open the Nessus interface in your browser, usually:

https://localhost:8834

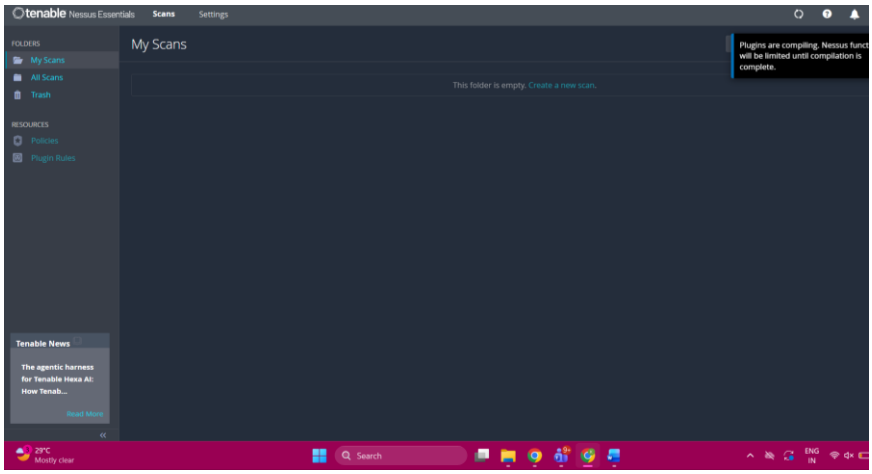


5. Complete the initial setup and activate Nessus Essentials using the activation code.

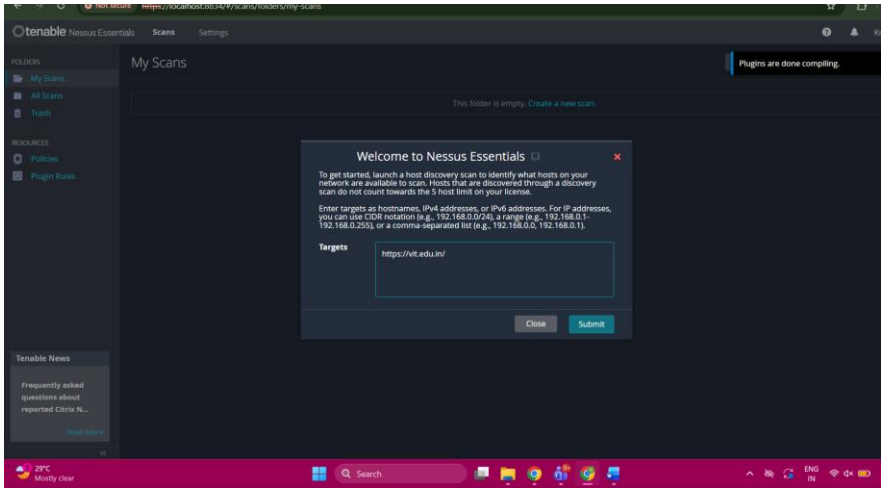
6. Wait for the plugins to download and compile.

Step 2: Create a New Scan

1. Log in to the Nessus dashboard.



2. Click New Scan.
3. Select Basic Network Scan.
4. Enter the scan name, for example: EXP 6 - Vulnerability Assessment.
5. Enter the target IP address of your lab machine.
6. Save the scan configuration.



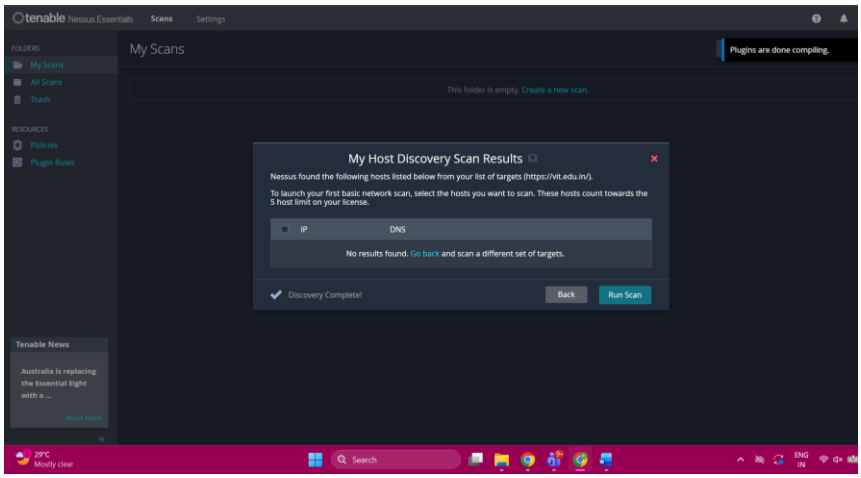
Step 3: Configure the Scan

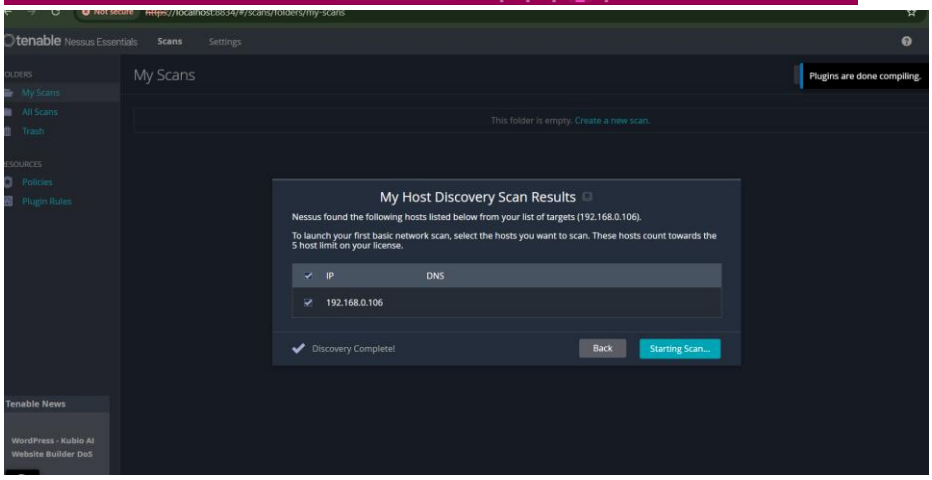
1. Open the Discovery tab and review the port-scanning settings.
2. Keep the default settings for a basic lab scan.

3. If credentials are provided for your lab system, configure them under the Credentials tab for a more detailed authenticated scan.
4. Save the configuration.

Step 4: Launch the Scan

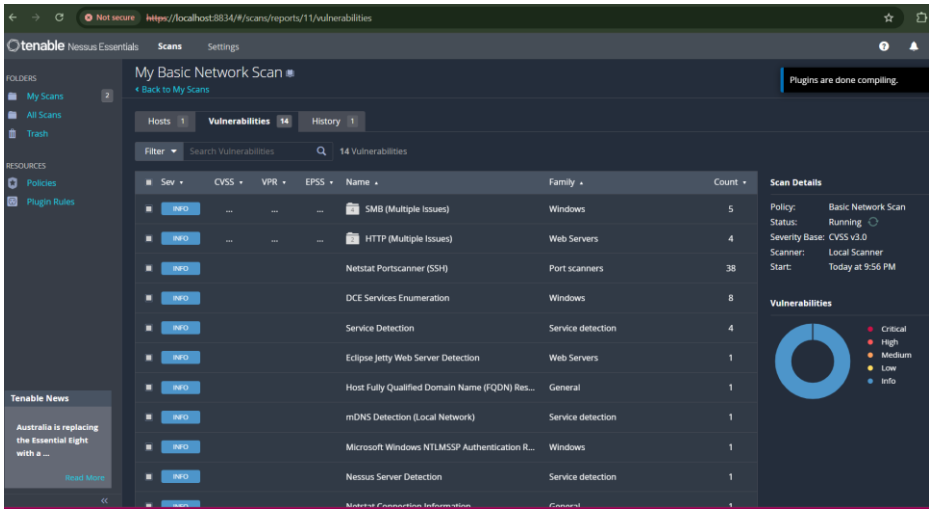
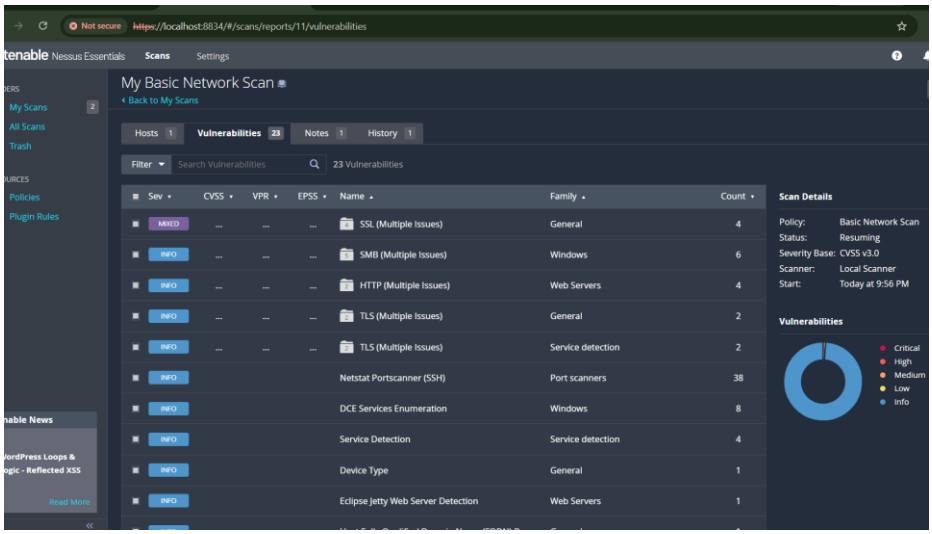
1. Click Launch to start the vulnerability scan.
2. Wait for Nessus to complete the scanning process.
3. Open the completed scan from the dashboard.





Step 5: Analyze the Results

1. View the vulnerabilities grouped by severity.
2. Select individual vulnerabilities to view their details.
3. Note the following information:
    - Vulnerability name
    - Severity
    - Host affected
    - Port and service
    - Description

	○ Suggested remediation  
Conclusion	Nessus helps identify vulnerabilities, outdated software, and security misconfigurations in a system. The experiment demonstrated how vulnerability scanning can support security assessment and help administrators prioritize remediation.